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Studierichting: Informatica
Oefeningenreeks 1C nr. 31.2

$$r = \sqrt{(-1)^2 + 1} = \sqrt{2}$$

$$\tan\alpha = \frac{-1}{1} = -1$$

$$\alpha = -\frac{\pi}{4} + \pi = \frac{3\pi}{4}$$

$$\left(\sqrt{2}\operatorname{cis}\left(\frac{3\pi}{4}\right)\right)^{10}$$

$$= (\sqrt{2})^{10}\operatorname{cis}\left(10 \cdot \frac{3\pi}{4}\right)$$

$$= 32\operatorname{cis}\left(\frac{15\pi}{2}\right)$$

$$\frac{15\pi}{2} - 2\pi \cdot 3 = \frac{15\pi}{2} - \frac{12\pi}{2} = \frac{3\pi}{2}$$

$$= 32\operatorname{cis}\left(\frac{3\pi}{2}\right)$$

$$= 32 \cdot (-i) = -32i$$

References